Analysing the Sales Trends of Various Food and Beverage Items in CCD, IITM Branch

A Proposal report for the BDM capstone Project

Submitted by

Renee Keerthana Paturi
22f2001379

IITM BS Degree Programme,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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<u>Declaration Statement</u>

I am working on a Project Titled "Analysing the Sales Trends of Various Food Items in CCD, IITM Branch". I extend my appreciation to **Café Coffee Day, IITM-Branch** for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: (**Digital Signature**)

Name: Renee Keerthana Paturi

Date: 30 December 2024

1 Executive Summary

The project focuses on one of the branches of CCDs present in IITM. It is a B2C in the segment of food and beverages, providing an ambient space to have discussions over coffee and snacks.

The major business issues that the organisation faces is restrictions from the authorities at IITM such as the Committee for Monitoring General Facilities for Students (CMGFS) which enable it to update its menu options offering variety to customers and gaining loyalty points along with accurate inventory of stocks to maximise profits. Since the last three years it has been unable to update its menu.
This branch already offers its eatables at prices subsidised at almost fifty percent compared to its other branches outside the campus.

Another issue that I have personally observed during my time in the campus CCD over the past few months is an issue with inventory stocking.

The second issue will be addressed by analysing the data via the ABC analysis approach, categorising items based on value and consumption, prioritising critical items and ensuring optimal stock levels.
To deal with the first issue, I shall get in touch with the authorities at CMFGS myself to understand the reasons why this supposed upgradation is put on hold.

The expected outcome helps the organisation reduce the money blockage in terms of inventory, which helps increase the profitability of the organisation.

2 Organisation Background

The company that I am working with is Café Coffee Day.
Founder: V.G. Siddhartha (1959–2019), a visionary entrepreneur.
Founded: 1996, with the first outlet opening in Bangalore.
Parent Company: Coffee Day Enterprises Ltd.
Inspiration: Siddhartha wanted to create a space where coffee lovers could enjoy premium coffee in a relaxed ambiance, inspired by global coffee culture.
CCD grew rapidly and became a favourite spot for casual meet-ups, business meetings, and youth hangouts.
The one currently open in IIT-M has existed since the last 10 years.

3 Problem Statement

3.1 Problem statement 1: Regulations from CMGFS (Committee for Monitoring General Facilities for Students) which is not providing them with a timely response of how to move forward since the last three years.

3.2 Problem statement 2: Proper inventory stocking which will maximise profits.

4 Background of the Problem

4.1 Lack of Permission from CMGFS: The committee is not responding with any answer as of yet because they want all items to be priced minimally for the students and perhaps they think that if they permit CCD to update their menu, they will automatically also increase the prices.

This is leading to a lack of response from their side leaving the discussion midair.

4.2 Inventory Stocking: Rough estimates for more items leads to incorrect assessment at times, which can be avoided. It would do the CCD some good to have an easy-to-use inventory management system that can optimise resource allocation.

5 Problem Solving Approach

To address the challenges being faced by CCD, I will first and foremost contact the authority at CMFGS to get a clear verdict from their end on what items do they prefer this specific branch to have in the college campus.
Secondly, a comprehensive data-driven problem solving and systematic approach is required to deal with the inventory stockage issue.

a. Methods Used

I shall be putting to use the ABC Analysis method, categorising inventory based on value and consumption prioritising critical items which will ensure optimal stock levels.

b. Intended Data Collection

- Sales Data: Total number of items sold for each menu item.

This data will be instrumental in quantifying consumption patterns, identifying popular items and optimising stock levels to minimise wastage and avoid stock-outs.

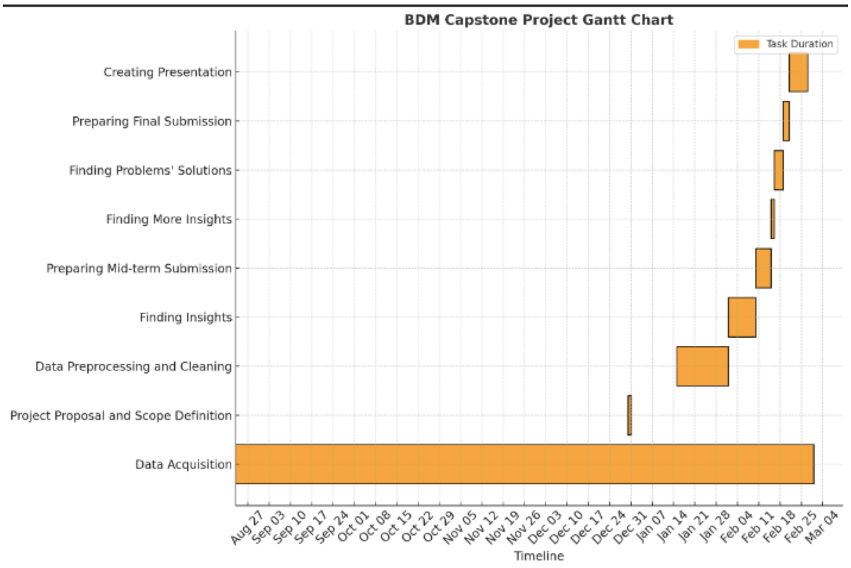
c. Analysis Tools

1. Google Sheets: This will be used as the central tool to manage inventory data effectively. It allows to input, organise, and store inventory information, facilitating data manipulation, calculations and generating tables for tracking stock levels and sales. The graphing and pivot table features help visualise trends and consumption patterns, enabling data summarisation and analysis. Pareto charts aid in identifying significant inventory items contributing to sales or wastage, assisting in prioritising efforts for inventory management.

2. Python and ML Tools: Python, a versatile programming language, offers numerous libraries and frameworks such as Pandas and NumPy which provide robust data manipulation, processing, and statistical analysis capabilities. With Python, large datasets can be handled, complex calculations can be performed and valuable insights from inventory data can be extracted.

The combination of these methods, data collection, and analysis tools will create a robust inventory management system. It will empower CCD-IITM branch to make data-driven decisions, reduce wastage, optimise stock levels and minimise operational costs effectively. The focus on accuracy and efficiency in inventory management will not only enhance customer satisfaction by ensuring product availability but also contribute to the CCD's long-term success and profitability in a competitive market.

6 Expected Timeline



The project is anticipated to be complete by the end of the January 2025 term, aligning with the project cycle of March 2025. The workflow follows the structure outlined in the Gantt Chart.

7 Expected Outcome

The expected outcome of this inventory optimisation project is to enhance the CCD's financial performance and operational efficiency by effectively tracking and quantifying inventory, thereby reducing wastage and stock-outs. This will result in minimised operational costs and improved customer satisfaction due to consistent product availability and quality.

Appendix

Location
CCD IIT-M Branch